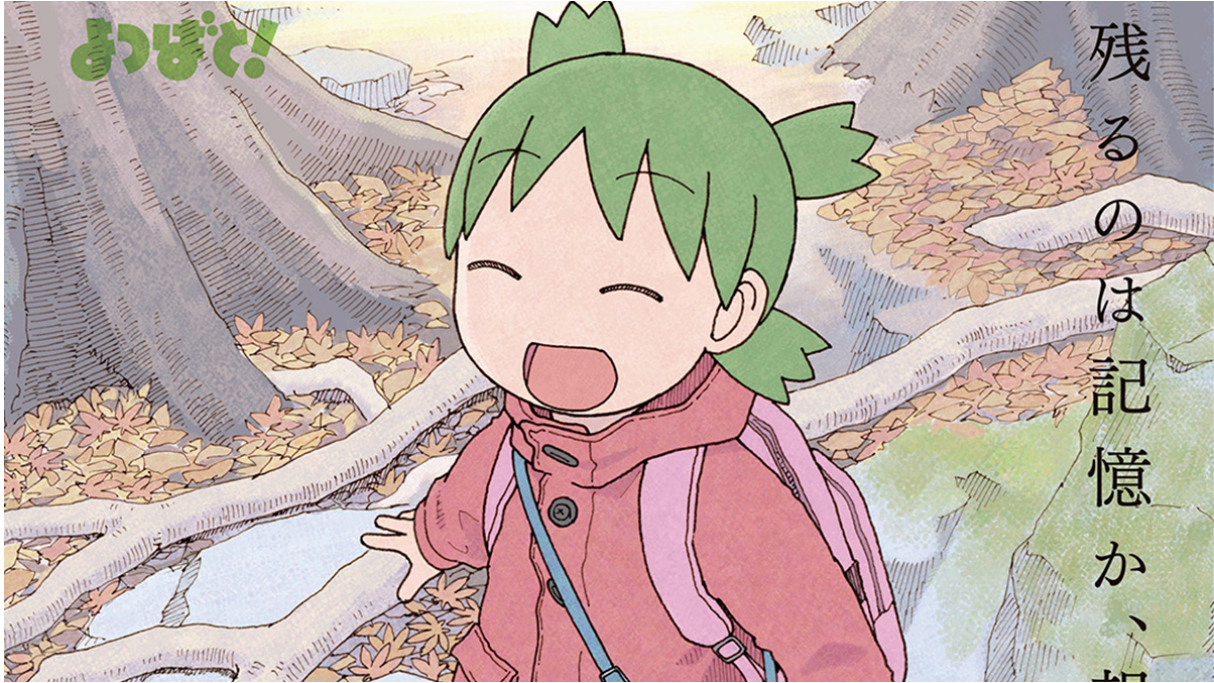


Linear Algebra Proofs

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Yotsuba from Yotsuba&!

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Exercise 1.1 Prove: If A and B are $n \times n$ matrices with the same n distinct eigenvalues, then A is similar to B .

Proof. Let A and B be $n \times n$ matrices with the same n distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ such that $A = PDP^{-1}$ and $B = QDQ^{-1}$ where $D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$.

$$\begin{aligned} A &= PDP^{-1} \\ \Rightarrow D &= P^{-1}AP \end{aligned}$$

$$\begin{aligned} B &= QDQ^{-1} \\ B &= Q(P^{-1}AP)Q^{-1} \\ B &= (QP^{-1})A(PQ^{-1}) \\ B &= (QP^{-1})A(QP^{-1})^{-1} \end{aligned}$$

□

Exercise 1.2 Prove the following. If A is a 2×2 matrix with a repeated eigenvalue λ , then A is diagonalizable $\Leftrightarrow A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$.

Proof. Let A be a 2×2 matrix with a repeated eigenvalue λ .

Prove: A is diagonalizable $\Rightarrow A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$.

- If A is diagonalizable, then the eigenspace associated with λ must have a dimension of 2.
- Thus, the reduced row echelon form of matrix $A - \lambda I_2$ needs two free variables.
- Thus:

$$\begin{aligned} A - \lambda I_2 &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \\ A - \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} &= \lambda I_2 \\ A &= \lambda I_2 \\ A &= \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ A &= \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \end{aligned}$$

Prove: $A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \Rightarrow A$ is diagonalizable.

- By definition, this is a diagonal matrix. Thus, it may be diagonalized as follows:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

By definition of biconditional, the original statement has been proved.

□